

Investment Appraisal

Payback Period:

- Payback Period is achieved when the cumulative cash flow becomes equal to the initial investment.
- Shorter the payback period, better the project.

Net Present Value (NPV)

NPV = Initial Investment **less** cumulative **PV** of all cash flows for 'n' years

$$\text{Present Value (PV)} = \text{FV} / (1 + r)^n$$

r = discount rate

n = valuation period in years

FV = Future Value

- A higher NPV is better. Higher the Discount rate, lower the NPV

IRR

The discount rate at which NPV is zero

Benefit Cost Ratio (BCR)

BCR < 1 : reject project

BCR > 1: accept the project

PERT / SD. (PERT can be used for both Time and Cost dimensions)

Formula	PERT (BETA Distribution)	Triangular Distribution
PERT Duration	$\frac{O + 4(ML) + P}{6}$	$\frac{O + ML + P}{3}$
Activity Variance σ^2	$\{(P - O) / 6\}^2$ (Standard Deviation squared)	$((P - O)^2 + (ML - O)*(ML - P))/18$
Activity Standard Deviation SD (σ)	$(P - O) / 6$	

Note: Default for PERT calculations is the BETA distribution formulas

Key

O – Optimistic
 ML- Most Likely
 P - Pessimistic

Confidence levels (Mean $\pm$ n σ)

1 Sigma (σ)	68.26%
2 σ	95.46%
3 σ	99.73%
6 σ	99.999%

To find range of estimate for Individual Activities: PERT duration for the activity $\pm$ SD

To find range of estimate at Project Level:

- Step 1: Add all the PERT durations in the critical path.
- Step 2: Calculate the Variance for each activity in the critical path.
- Step 3: Add all the Variances.
- Step 4: Take square root of the sum of all the Variances which gives the SD.
- Step 5: Project duration range estimate is total project PERT duration (step 1) $\pm$ SD (step 4).

Earned Value Formulas

$$SV = EV - PV$$

$$CV = EV - AC$$

$$SPI = EV / PV$$

$$CPI = EV / AC$$

Forecasting EAC

- **EAC = BAC / cumulative CPI** - when ETC work i.e. remaining work is predicted to be performed at the cumulative CPI. This assumes the to date CPI will continue in future.
- **EAC = AC + (BAC - EV)** - when remaining work is predicted to be performed exactly as per the original budget. Assumes any variances till date – both favorable or unfavorable - will not continue in future.
- **EAC = AC + bottom-up ETC** - when totally new detailed bottom-up estimates are developed for the remaining work.
- **EAC = AC + [(BAC - EV) / (cumulative CPI x cumulative SPI)]** – when both cost and schedule performance indices are considered for performing remaining work. Most useful when project schedule impacts ETC effort. CV is assumed to be negative.

TCPI: To-Complete Performance Index

$$\frac{\text{Work Remaining (BAC - EV)}}{\text{Funds Remaining (BAC - EV) or (EAC - AC)}}$$

TCPI based on BAC

$$TCPI = \frac{BAC - EV}{BAC - AC}$$

TCPI based on EAC

$$TCPI = \frac{BAC - EV}{EAC - AC}$$

Earned Value Acronyms & formulas

Acronym	Term	Definition
PV	Planned Value	Planned cost or value of the work to be done till this point in time.
EV	Earned Value	The value of the work accomplished till this point in time. Cost is as per the original budget.
AC	Actual Cost	The costs actually incurred to complete the work till this point in time.
BAC	Budget At Completion	The total planned value or budget for completing the entire project.
SV	Schedule Variance	Difference between the scheduled completion and actual completion of an activity or group of activities. Negative SV - is behind schedule. Positive SV - is ahead of schedule.
CV	Cost Variance	Difference between the budgeted cost of completing an activity/group of activities and the actual budget spent for it. Negative CV: is over budget. Positive CV - is under budget.
SPI	Schedule Performance Index	The measure of efficiency in managing the project's schedule. SPI > 1 is good (ahead of schedule) = 1 on target < 1 poor (behind schedule)
CPI	Cost Performance Index	The measure of efficiency in managing the projects budget. CPI > 1 is good (under budget) = 1 is on target < 1 is poor (over budget)
EAC	Estimate At Completion	Prediction of what project will cost when completed. EAC is calculated using different formulas for different possible conditions.

Important Formulas for the PMP® exam

Acronym	Term	Definition
ETC	Estimate To Complete	How much more we expect project to cost from this point in time.
VAC	Variance At Completion	How much under budget or over budget we expect the project to be once it is completed.
TCPI	To-Complete Performance Index	The cost performance needed in project for remaining work to stay within the planned budget (BAC) or the estimate at completion (EAC). Is the ratio of “work remaining” to “funds remaining”.

Some Contractual Terms

Arbitration	Settling a dispute out of court using an independent third party. The arbitrator must be agreed upon and accepted by both parties.
Breach of contract	Violating or breaking of a legal obligation. Is a serious condition. Buyer should always issue letter to contractor notifying the breach.
Contract	A written or oral agreement made by one party to another that has legal obligations on both parties.
Condition	A term of fundamental importance in the contract. Breach of this condition can cause the contract to be terminated.
Design specifications	A detailed description of the physical characteristics describing and specifying what is to be done.
Force Majeure	Used in contracts to free both parties from liabilities arising from events beyond their control e. g. strikes, war, floods, earthquake etc. Common response is for buyer to extend the time.
Good faith	Transparency and fair dealing between all parties.
Infringement	Violation of a legally recognized right.
Indemnity	A payment or compensation as protection against any future loss. It is an obligation made by one party to reimburse another party for losses that have occurred or that may occur in future.
Liquidation damages	Reasonable damages to be paid by the contractor to the owner due to failure to complete the specified work as per the contract terms.
Negligence	Not acting in a reasonably accepted manner.
Non-compete clause	The contractor is not allowed to work for a competitor for a given time.
Non-disclosure / confidentiality clause	A restriction on the contractor from disclosing some proprietary knowledge gained in doing the work.
Penalty clause	An agreement made in financial terms to be paid by the contractor for not performing as per the contract terms.
Performance specifications	The measurable capabilities that the product should achieve in terms of operational characteristics. They must be met by the contractor.
Privity of contract	A mutual relationship that exists between a buyer and seller. The contract cannot give rights or impose obligation on any person / party / sub-contractor except the parties that have signed the contract.
Screening system	A process used to determine if a contractor has the minimum qualifications to bid.
Sole source	The seller is the only available source for the procurement.
Waiver	Giving up of a legal right or privilege voluntarily
Warranty	A written, verbal or implied promise assuring that a specified provision in the contract is true. Provides protection to the buyer against breakdowns and major repairs.

Types of Estimates

Rough Order of Magnitude (ROM) estimate - Initiating phase: +/- 50%

Budget Estimate - early Planning phase: -15% to +25%

Definitive or detailed estimate - Planning phase: +10% to -10%

Communication channels

Communication channels between people = $\{n(n-1)\} / 2$ n = total number of persons

Procurement

Fixed Price Incentive Fee (FPIF) contract

Point of Total Assumption (PTA): The point at which additional cost overruns have to be fully borne by contractor. Costs above PTA are assumed to be the result of mismanagement. PTA is only applicable in FPIF contracts.

$$PTA = \{ (\text{Ceiling price} - \text{Target price}) / \text{Buyer's share ratio} \} + \text{Target cost}$$

- **Price:** The amount charged to buyer by seller (contractor)
- **Target cost:** Expected cost for doing the work at time of signing the contract
- **Target fee:** Sellers planned profit margin or fee for doing the work. Will be increased / decreased using the Share ratio based on performance
- **Target Price:** Target cost + target fee
- **Share ratio:** Ratio by which Buyer/Seller will share cost savings and cost overruns
- **Ceiling Price:** The maximum amount the buyer will pay for the contract irrespective of the costs.
- **Actual Cost:** Costs that actually incurred at end of contract

Cost Plus Incentive Fee (CPIF) contract: CPIF includes all of the above terms except Ceiling Price and Point of Total Assumption (PTA) . Instead CPIF has a Minimum fee and a Maximum fee:

- **Minimum Fee:** Minimum assured fee buyer will pay to contractor
- **Maximum Fee:** Maximum fee that buyer will pay to contractor

Quality

– The degree to which a set of inherent characteristics fulfils (project) requirements

Quality Tools

Pareto Chart

- 80/20 rule. Is a histogram ranking no. of defects in order of frequency or importance
- 80% quality problems due to 20% causes

Control chart

- Used to decide whether the product or service is in control or out of control
- Identifies special or assignable causes
- Has a mean or center line, an upper control limit (UCL) and a lower control limit (LCL)
- Does not show causes for deviation or provide solutions

Cause-and-Effect diagram (Fishbone or Ishikawa diagram)

- Graphical technique that helps team to group ideas and identify the causes of a problem
- Breaks down problem for analysis
- Shows how different variables may be linked to the effect (problem)

Sampling

- **Attribute sampling:** checks that the result either conforms or does not conform – pass or fail
- **Variable sampling:** checks the degree to which the result conforms – acceptable within a tolerance level

Risk Response strategies

Risk	Response Strategy	Effect of Response
Threats i.e. Negative risks	Avoid	Remove root cause. Project management plan changed to completely remove threat e.g. extend schedule, reduce scope by removing work package or activity, remove a team member, etc.
"	Transfer	Third party made financially responsible for negative impact and ownership of response. e.g. insurance, contracts, warranties etc. Transfer does not eliminate the risk.
"	Mitigate	Probability or Impact or both to acceptable threshold limits e.g. make a prototype, improve skills through training
"	Accept	No change made in the project management plan to deal with risk or unable to select suitable response. Passive acceptance – no action taken Active acceptance - Contingency reserve commonly kept – time, money or resources to deal with the risk
Opportunities i.e. Positive risks	Exploit	Change plans to remove uncertainty to ensure the opportunity occurs
"	Share	Allocate some or all ownership of opportunity to third party as they are better equipped e.g. joint ventures, teams, partnerships
"	Enhance	Increase positive Impact or Probability or both of opportunity e.g. add more resources to finish early
"	Accept	Accept the risk if it occurs. Not actively pursued